Predicting Market Volatility and Short-Term Trading Signals from Reddit's WallStreetBets

UE24CS352A Machine Learning - Mini-Project write-up | Team: Prithvi Aithal (PES1UG24CS909), Sagar A Walikar (PES1UG24CS396) | Repository: github.com/Prithvi-Aithal/wsb-market-prediction (private)

1. Problem statement

Retail-investor communities such as r/WallStreetBets (WSB) moved prices during the January 2021 GameStop short squeeze. We ask whether the **posts themselves** contain a usable signal about (a) the **direction** of the next price move (will open(next day) exceed open(this day)?), (b) **volatility** (will the move be unusually large?), and (c) whether such predictions can drive a simple short-term trading strategy. We re-implement the CS229 report's approach and extend it with hyper-parameter tuning, stricter baselines, a leakage-free evaluation and a back-test.

2. Dataset

Posts: Kaggle *gpreda/reddit-wallstreetsbets-posts* - 53,187 posts (Sep 2020 - Aug 2021) with title, score, id, url, comment count, creation time, body, timestamp. **Prices:** daily OHLC from Yahoo Finance (*yfinance*). Posts mentioning each stock: GME 14,619, AMC 5,310, TSLA 1,232, FB 251 (ticker or company name). Each post becomes one example, as in the report.

3. Approach

Pre-processing: drop NaN titles, lowercase, remove URLs, handles, special and single characters, collapse spaces. **Features (7)**: post_score, total_comms_num, len_of_post, positive/neutral/negative sentiment (VADER), bias term; count features are log1p-transformed and standardised on the training set. **Labels**: *direction* = open(k+1) > open(k); *volatility* = |open(k+1)/open(k)-1| above the training-set median (the report's 'significant shift'). Two timing conventions: *paper* (k = post's trading day) and *tradable* (k = first open after the post, so no look-ahead). **Models**: (1) logistic regression solved with **Newton's method implemented from scratch** (small L2 term keeps the Hessian invertible); (2) 3x256 ReLU network, sigmoid output, Adam (lr 1e-3), 30 epochs, batch 16 (64 for GME/TSLA), best epoch by validation accuracy; (3) uniform-random and majority-class baselines. **Tuning (our extension)**: logistic-regression L2 in {1e-6..10} chosen by validation log-loss; network random search (6 configs: depth/width, learning rate 1e-4..3e-3, weight decay 1e-4..1, batch 32-128) chosen by validation accuracy, then re-trained with 5 seeds. **Evaluation**: chronological 80/10/10 train/val/test split so the test set is strictly in the future.

4. Implementation overview

Python; *src/wsb_pipeline.py* (data, features, labels, splits), *src/models.py* (Newton logistic regression, network), *src/run_experiments.py* (experiments, tuning, baselines, back-test, figures), *src/leakage_check.py* (shuffled vs chronological split), *src/demo.py* (live scoring of a post, and a *--quick* mode that runs the whole pipeline live on one stock), *tests/* (unit tests).
Back-test: for every trading day, majority vote of the day's per-post predictions -> long (open to next open) or flat, compared with going long on every day that had posts.

5. Results (held-out chronological test set, accuracy)

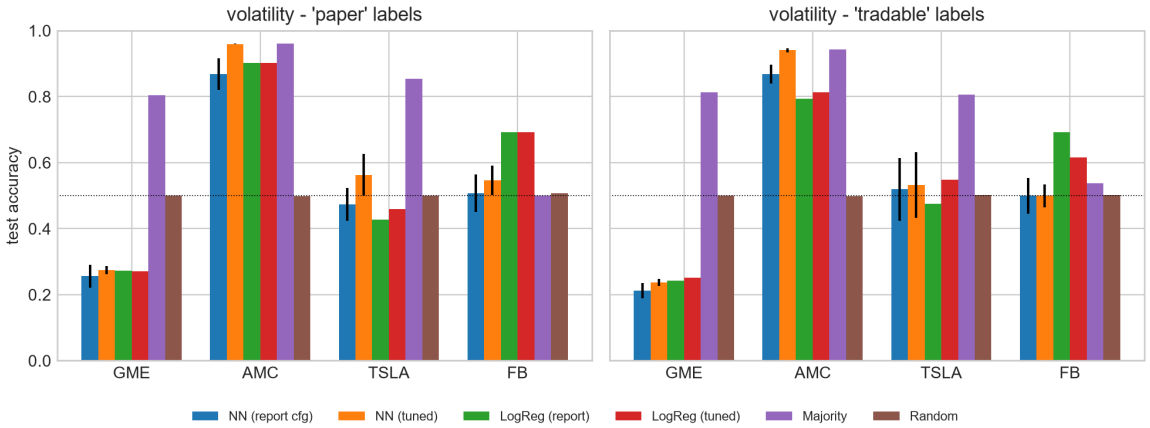
Direction task, 'tradable' labels

Stock	n test	NN (report)	NN (tuned)	LogReg (report)	LogReg (tuned)	Majority	Random
GME	1462	0.569	0.569	0.567	0.567	0.567	0.500
AMC	531	0.350	0.353	0.411	0.388	0.653	0.500
TSLA	124	0.563	0.494	0.573	0.589	0.556	0.502
FB	26	0.500	0.469	0.462	0.538	0.577	0.495

Volatility task, 'paper' labels

Stock	n test	NN (report)	NN (tuned)	LogReg (report)	LogReg (tuned)	Majority	Random
GME	1462	0.256	0.275	0.272	0.271	0.805	0.500
AMC	531	0.869	0.960	0.902	0.902	0.960	0.499
TSLA	124	0.474	0.563	0.427	0.460	0.855	0.500
FB	26	0.508	0.546	0.692	0.692	0.500	0.507

Leakage check (report's direction task, report configuration): with a *shuffled* split the network scores GME 0.77, AMC 0.78, TSLA 0.53, FB 0.68; on our *chronological* split it scores GME 0.57, AMC 0.35, TSLA 0.60, FB 0.47. Same-day posts share a label, so shuffling leaks the answer.



6. Conclusions

(i) **Honest evaluation matters.** We could not reproduce the report's high accuracies (GME 0.84, AMC 0.82) on a time-ordered split; a shuffled split inflates AMC and FB, which suggests leakage is a likely cause (we cannot verify the report's exact split). On a future hold-out no direction model beats the majority-class baseline by more than the test-set standard error (see table; AMC is far below it), so we cannot claim WSB post features predict next-day direction from these 7 features. (ii) **Tuning helps only modestly:** for AMC volatility the tuned network improves from 0.87 to 0.96 (only matching the trivial majority baseline 0.96), and for TSLA from 0.47 to 0.56 (still far below its majority baseline 0.85); for the rest it mostly selects the trivial 'predict the majority' solution. (iii) The back-test confirms it: the strategy stays flat on GME/AMC/TSLA (it never predicts 'up') and slightly under-performs always-long on FB. (iv) Distribution shift is the main obstacle: post volume peaks during the Jan-Feb 2021 squeeze, so train and test regimes differ; TSLA and FB test sets are tiny (124 and 26 posts, std. error 5-10 pp), so their differences are not significant. **Future work:** richer text features (BERT embeddings), day-level aggregation, more data and walk-forward validation, probabilistic calibration, and a transaction-cost-aware strategy.